THE FOOD SYSTEM OF BAHRAIN:

ANALYZING RISKS AND POSSIBILITIES

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Abstract

The country of Bahrain has experienced unprecedented growth in the past decades, resulting in a unique situation in the development of their food industry and a marked reduction in the country's ability to produce food domestically due to geographic barriers and a shift away from the traditional fishing industry in favor of international trade. Complicating matters, Bahrain's population has increased dramatically since industrialization and the country's focus on oil and aluminum refining does not offer a stable economic future. This situation, however, exposes Bahrain's food security to political risk because the population of Bahrain at this point cannot be supported through domestic production alone. The issue of food sovereignty also plays a role in Bahrain's food system, as the Western diet becomes more pervasive in the country. Although many other countries in the Gulf region face similar risks, Bahrain faces the most limits in developing domestic food production due to its small land area and very limited water supply, making it an interesting case study for the problem of food import reliance.

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Introduction

Agriculture can be referred to as one of the most basic economic necessities for any society. Not only is it necessary for a society to be fed, agriculture can also create opportunities for growth through trade. In the case of Bahrain, a country that consists of a series of islands in the Persian Gulf, a mix of small land area, hot temperatures, lack of water, unprecedented population density, and changing food consumption trends create an interesting case study to acknowledge the issue of food import reliance in the Gulf region. Throughout history, food trade has been a necessity in Bahrain, but in the 21st century the need to adequately feed and offer nutritionally beneficial food to residents of the islands of Bahrain presents a need to be more aware of possible risks to food and agriculture in Bahrain. In order to better understand Bahrain's food system, its history, parts, and modern-day realities will be examined, especially in light of the World Economic Forum's 2017 report and events current to the time of publishing this paper.

Research for this report was conducted by collecting recent information about trade relationships that Bahrain maintains.This information is provided by sources such as the UN Food and Agriculture Organization, a research and support group that aims to solve problems of governance and policy in food and agriculture worldwide, as well as similar organizations. This report also draws on historical analyses of agriculture in Bahrain, alongside reports that characterize worldwide upcoming risks in food trade looking towards the year 2050 and subsequent global population growth.

The report starts with a summary of agriculture in Bahrain through sections on fisheries and traditional fishing practices, small-scale agriculture, and historical trade in the region as it pertains to Bahrain. Then, the report will continue by focusing on population growth in the past 20 years alongside industrialization, modern food trade, and changes in Bahraini food culture and consumption habits in the last 20 years. Finally, certain risks will be identified and evaluated such as those presented by trade dependence, changes in consumption, and a deficiency of food sovereignty, followed by predictions offered by the World Economic Forum for changes in the global food supply in the next 30 years.

The State of Agriculture and Food Trade in Bahrain

Food production in Bahrain consists historically and today of three main parts, domestic fishery activities, agriculture, and food imports. This section will describe those parts in detail, addressing historical changes and current situations.

Fisheries

The Bahrain Islands have long participated in fishing both for food consumption and to supply the profitable pearl industry. Bahrain now no longer participates in fishery activity on large scales, and is unable to meet domestic fish product needs through domestic production alone.

Prior to industrialization in the mid 20th century, most Bahraini men participated in fishing at some point, with many working as pearl divers until the decline of the natural pearl industry in the 1950s, or as fishermen until the economy of Bahrain shifted away from agriculture as the oil and aluminum industry grew.[1] As industry grew in the Persian gulf, pollution [2] and overfishing [3] played a role in diminishing the magnitude of fishing in Bahrain to the point that a rapidly growing and traditionally important shrimping industry in the 1960s reduced enough to be considered economically insignificant by the UN FAO by 2018.

At the present, fishery operations in Bahrain are artisanal. That is, they are operated to supply some fish products domestically and on small scales, but are not capable of meeting the full demand for fishery products in Bahrain.[5] Accordingly, Bahrain has become primarily an importer of fish and fishery products, with the total value of imports at USD 38.9 million in 2017[2] and the total value of fish, crustacean, mollusk, and aquatic invertebrate imports as reported by the UN Comtrade database at USD 64.82 million in 2018.[4] These figures agree with the trend of increasing costs of imports since 1999, and may be representative of increasing demand and decreasing domestic fishery production. It should be noted that the values of imports and exports of fishery products to Bahrain have been more or less equal for decades, but that a large amount of exports may be classified as “re-exports” or products imported in order to be further processed into food products.[2] Furthermore, a large portion of fish products in recent

years have originated outside of Bahrain's territorial waters,[5] a fact that reflects environmental and regulatory issues and therefore stimulates an increased demand for imported fish products.

Figure A: Political map of Bahrain with inset map of surrounding areas

Agriculture

Bahrain has historically participated in agriculture in order to produce date palm fruits for export. Since the 1980s, however, agricultural production has decreased as industry developed to

focus more on oil and aluminum production, while the national supply of fresh water, a necessity for agricultural activity, has become more scarce.

Agriculture on the Bahrain Islands takes place on a number of small inherited plots, with the majority being owned by the Al-Khalifa ruling family.[2] This system has been in place for decades, but does not constitute a large enough portion of consumption to impactfully be considered a form of production. In fact, date farming has historically made up the majority of agricultural activity in Bahrain. Today, date farming in Bahrain is artisanal in nature, taking place on a small-scale and being perceived as secondary in importance to food imports. Another part of agricultural production, livestock production is limited and faces restrictions, such as the necessity for feed to be imported, and underdeveloped animal husbandry techniques compared to other countries.[6]

Subsidies have been a primary tool for the government to maintain farming as a viable way of life in Bahrain, but corruption and inefficient bureaucratic systems have stymied those efforts, leading to a decrease in the number of farmers from 15,000 to 650 between 2002 and 2016.[7] As a result, many farmers forgo land ownership and sell their plots.[7] In 2015, the National Initiative for Agricultural Development initiated the King Hamad prize to reward work in agriculture at several levels, including best Bahraini farmer, best agricultural project, and best agricultural research and studies.[8] Among the research categories suggested by the National Initiative for Agricultural Development are methods for determining "the content of heavy metal and harmful material in locally produced and imported fruits and vegetables." This is an initiative that takes into account the idea of food sovereignty, an issue that Bahrain, as a primary food importing country, must consider when determining the quality of food being consumed.

Access to water is another issue that hinders further development of the agricultural industry, and is recognized by the King Hamad prize as an important area for innovation. In 2003, irrigation and livestock accounted for 45% of water usage,[6] and could be considered domestically as an expenditure without sufficient value when compared to Bahrain's ability to import agricultural products. Groundwater reserves have become contaminated by seawater, further limiting the availability of this resource.[6]

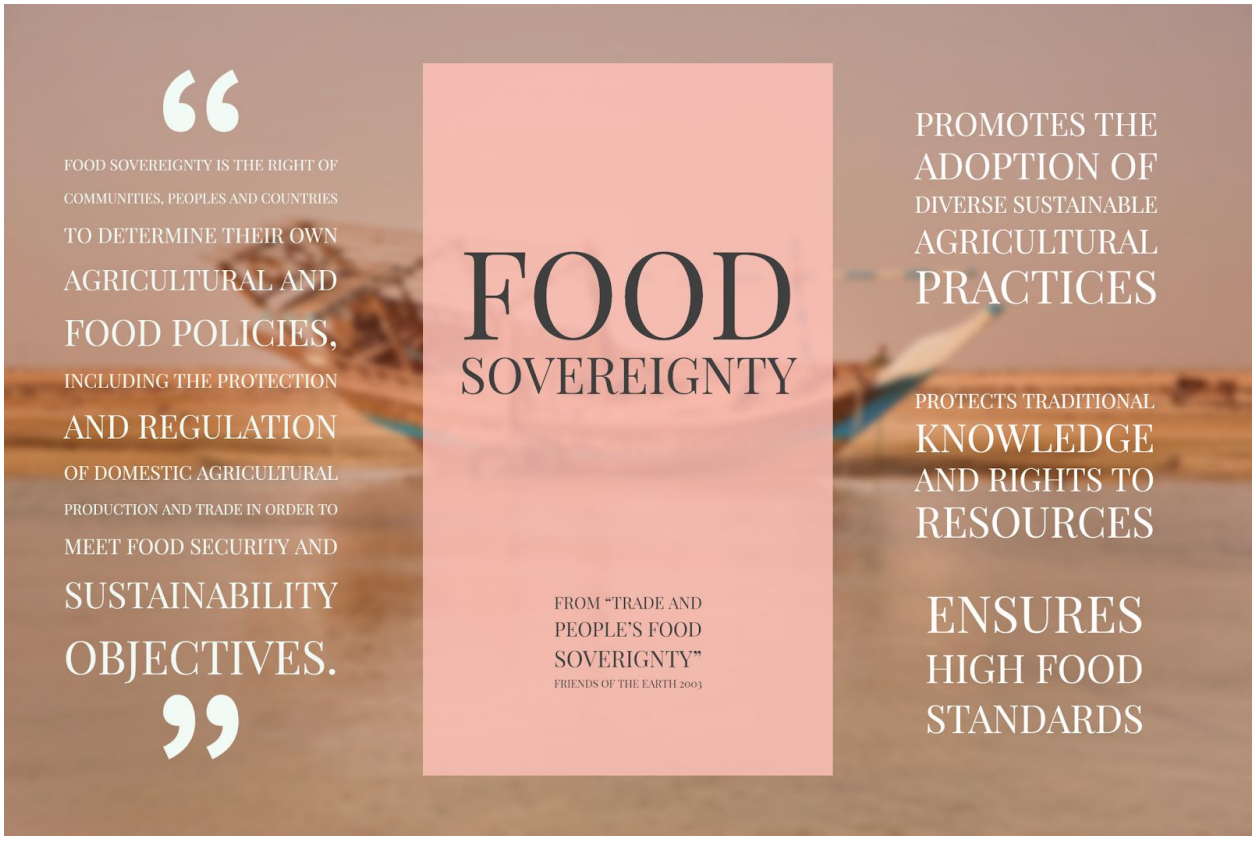

Figure B: Informational graphic about Food Sovereignty

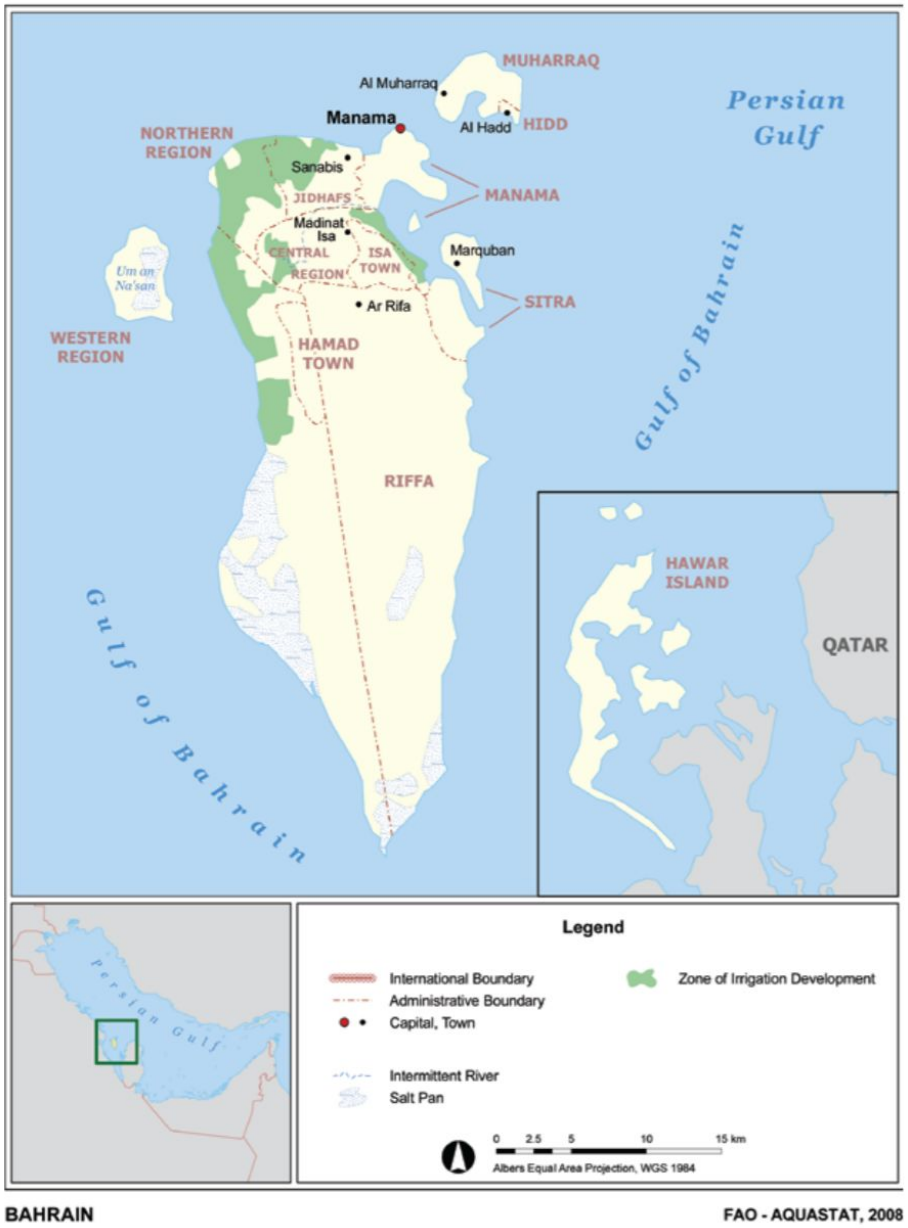

Figure C: Map of Bahrain showing administrative boundaries and areas designated for agriculture (irrigation development in green)

Trade and Trade Partners

Trade on the island of Bahrain is widely considered to have taken place as early as the 3rd century BCE and consisted of a connection between Mesopotamia and the Indus Valley. Up until the mid-20th century, Bahrain continued to be a substantial center of trade, largely based around the pearl industry, offering cultural and material exchange for Bahrainis. Since then, trade has continued, based on oil and aluminum industries in Bahrain and imports of materials not found in the island archipelago.

Major importers for Bahrain are China, Japan, the UK, Turkey, and the US.[9] Major export destinations from Bahrain are the US, South Africa, South Korea, Japan, and India.[9] Trade with China focuses on importing machinery and metal materials. Trade with Japan focuses on importing vehicles and exporting refined petroleum and aluminum. Trade with the UK focuses on importing machinery and vehicles, and trade with the US focuses on exporting aluminum products and importing vehicles and machinery.

From a strictly monetary perspective, none of these items are substantial to the GDP of any country involved, but when focusing on food trade, raw value is not the most important factor to consider. Food is not very expensive, especially when compared to the price and consumption of oil, so countries determined to be less economically influential in economic trade can be much more influential in food trade. Bahrain's food imports, though, are noticeably diversified. No one country supplies the majority of food for Bahrain

Developments Over the Last 20 Years

Development in the Gulf region of the Middle East has accelerated since the discovery of oil. In Bahrain, this era of unprecedented growth has transformed the country in terms of population change, industrialization, new agricultural activities, and a dramatic change in food consumption. In many countries, changes in any one of these areas would constitute a significant development for the country, but in Bahrain, due to its small land area, these changes have proven to be wholly transformative. In a period of a few decades, Bahrain reconstructed its economy from depending on pearls to depending on oil and aluminum products. Despite recent

discoveries of additional oil reserves, Bahrain's small territory does not allow the country to indefinitely continue sourcing crude oil for refinement from domestic sources. In effect, Bahrain must find its own solution to a common issue in the Gulf region: economic diversification, especially to ensure a secure food supply in the future.

Population Change

From 1950 to 2020, the population of Bahrain has increased from about 115,000 to over 1.7 million people.[10] Today 46% of the population is considered to be Bahraini citizens.[11] Interestingly, about 65% of the population as a whole are males, a factor likely caused by migrational work patterns.[11] The population of Bahrain is predicted by the UN to increase by 600,000 individuals by 2050.[10] Although this type of growth is not necessarily cause for concern, in terms of the food supply, Bahrain's reliance on food imports means that the country relies on trade partners to implement agricultural practices that ensure long-term food stability. As the population has grown, simultaneous poor support for small-scale farming has homogenized the food supply in Bahrain, leaving little domestic ability for food production. This can been seen through the statistic that Bahrain's rural population decreased from about 17% in 1960 to less than 11% in 2018[12], a factor that the United Nations Food and Agricultural Programme (UN FAO) suggests correlates with food production ability.[13]

The Economy

In the mid 20th century, Japanese technological advances created the artificial pearl industry and crashed the Bahraini pearl industry. Now, petroleum is the cornerstone of Bahrain's trade economy with large contributions also coming from aluminum refining and the construction industry. Food production as an economic activity has decreased consistently despite a short period of growth in the 1980s, leaving the economy largely dependent on petroleum. Recent efforts to continue diversifying the economy have met relative success compared to other countries in the Gulf region, but the country's dependency on oil production is still evident. In 2018, due to decreasing oil prices, Bahrain accepted an economic bailout from

Saudi Arabic, Kuwait, and the United Arab Emirates, a move that was likely motivated by interests in maintaining political stability in the region and preserving the strong relationship between Saudi Arabic and Bahrain.[12] The Bahraini Government's approach to increasing diversification, Economic Vision 2030, published in 2008,[16] gives little systematic guidance for a complete transformation of the economy, but rather continues to place hope in high impact sectors such as investing, technology, and entrepreneurship. Perhaps counterproductively, the Bahraini Government takes a leading role in entering new industries, as seen in the state-owned oil and aluminum industries, which may stymie private initiatives for diversification. Food production enterprise, though, is not only an existing need for Bahrain, but also an opportunity for diversification given issues such as water scarcity are solved.

Agriculture and Food Trade

Agriculture in Bahrain increased output in the decade up to 1980, after which it declined again as the petroleum market stabilized. Since about 2008, however, concerted efforts have paid off in increasing food production.[17] Little data is available about the nature of the increase in food production, but the UN FAO makes it clear that, from 2000 to 2008, fruit and dates were the two most produced commodities on average, but from 2008 to 2018, vegetable production overtook date production while simultaneously decreasing from an average of about 14,000 tonnes to an average of about 11,600 tonnes.[18] These numbers indicate that the country on a whole was reassessing the importance of date production in relation to vegetables, a food product with higher domestic demand, and, surprisingly, lower water consumption.

In terms of trade, despite increases in domestic production, the evidence still suggests that Bahrain is disproportionately dependent on imports for food supply. Because of this, issues can arise in food sovereignty, as it is more difficult to impose strict quality standards on imports. *Figure D* below details Bahraini food import requirements as described by the USDA in 2010. While the requirements seem standard, food labs testing samples of imports would be required to ensure the products are not contaminated with harmful substances.

Section I. List of All Export Certificates Required By Government (Matrix) :

Products	Title of Certificate	Attestation Required on Certificate	Purpose	Request-ing Ministry
Dry pea, lentil, Rice, seeds, fresh fruits & vegetables, etc.	Phytosanitary Certificate	This is to certify that the plant(s) products described above: -have been inspected according to appropriate procedures, and -are considered to be free from quarantine pests and particularly free from other injurious pests, and -are considered to conform to Phytosanitary regulations of the importing country	Pest and insect free confirmation	Public Health Directorate Ministry of Health, Agriculture Department
All products	Country of Origin Certificate	The goods are of "xxx" origin	Country of origin confirmation	Customs
All products Except meat And poultry	Health Certificate	The goods have been prepared and shipped under sanitary conditions, do not contain harmful ingredients, and are fit for human consumption.	Health Certificate	Public Health Directorate Ministry of Health

All Products produced in Europe	Radioactivity Certificate 1/	The goods have been tested for radioactivity and found to be safe for human consumption Cs-134: < 10 Bp/kg Cs-137: < 10 Bp/kg	Health Certificate for products coming from Europe	Public Health Directorate
Meat & Poultry and other products containing animal ingredients except pork.	Halal Slaughter Certificate	Animals were slaughtered in accordance to Islamic/Sharia laws & requirements	Product produced in accordance with Sharia Law confirmation	Public Health Directorate

1/ required for products originating from Europe only

Section II. Purpose of Specific Export Certificate(s)

See Section I above.

Section III. Specific Attestations Required on Export Certificate(s)

a. Halal Slaughter Certificates have to be legalized by the Bahraini Embassy in the exporting country. Embassy from any other Arab country will suffice in the absence of Bahraini Embassy.
b. If exporter fails to legalize the Halal Slaughter Certificate, the Bahrain importer may do the notarization locally in Bahrain

Section IV. Government Certificate's Legal Entry Requirements

- Original certificates must accompany shipment at the time of entry. Sometimes importers are allowed to have the documents sent to their banks.
- Certificates are used for only the accompanying shipment since it carries the details of the shipments.
- U.S. State issued export or health certificates are acceptable
- Modification to any certificate is not permitted once issued and shipped to accompany the product.

Section V. Other Certification/Accreditation Requirements

Original invoice, Bill of Lading and Packing List are required to accompany the above-referenced certificates to complete the list of documents required by Custom Authorities for release of the shipment.

Appendix I. Electronic Copy or Outline of Each Export Certificate

The following are "samples" of certificates outlined in section I and accepted by Bahrain officials. The standardized export health certificates issued by FSIS and APHIS are not repeated here.

Note: At present Bahrain does not require a specific format for any certificate. Bahrain seeks only the information provided therein.

Figure D: Bahraini food import requirements as detailed by the USDA in 2010

A Changing Food Culture

An increasingly globalized food culture, urbanization, increasing buying power in the Middle East, and discounting drought tolerant and local crops for local markets has led to a Westernization of diets in the Middle East.[20]

Bahrain's instability caused by a reliance on food imports is not an individual phenomenon, but can be seen as a trend across the Arabian Gulf. Among all Arab countries, Gulf States, in particular, are seen as epicenters of the change due to rapidly increasing buying power on the international market and high standards of living. This change, however, has presented unfavorable results in terms of public nutrition. Non-communicable diseases like obesity and diabetes are shown to increase steadily as a Western style diet is adopted.[19]

Another consequence of this change, though, is a move towards crops that are less adapted to growing unprotected in the hot, humid environment of the Gulf region. Drought tolerant crops such as chickpeas, fava beans, Molokhia (*Corchorus olitorius*), cowpeas, sorghum, and local heirlooms aren't considered staples of the western diet, but can constitute an important part of food production independence for Bahrain if implemented into the daily diet. Including more traditional regional crops and crop varieties in the diet of Bahrainis would also help in maintaining the food sovereignty of the region, allowing those in the Arabic Gulf to interact culturally and sustainably with food. Alternatively, through plant breeding and genetically guided crop development, future crop varieties may be more suitable to the challenging agricultural environment of the Middle East.

Altogether, the effect of increasingly Western diets in the Middle East can be seen as detrimental, not only from a public health standpoint, but from the damage done to regional food sovereignty, and to the environment through growing crops that cannot be grown efficiently in the local environment, albeit currently on a small overall scale.

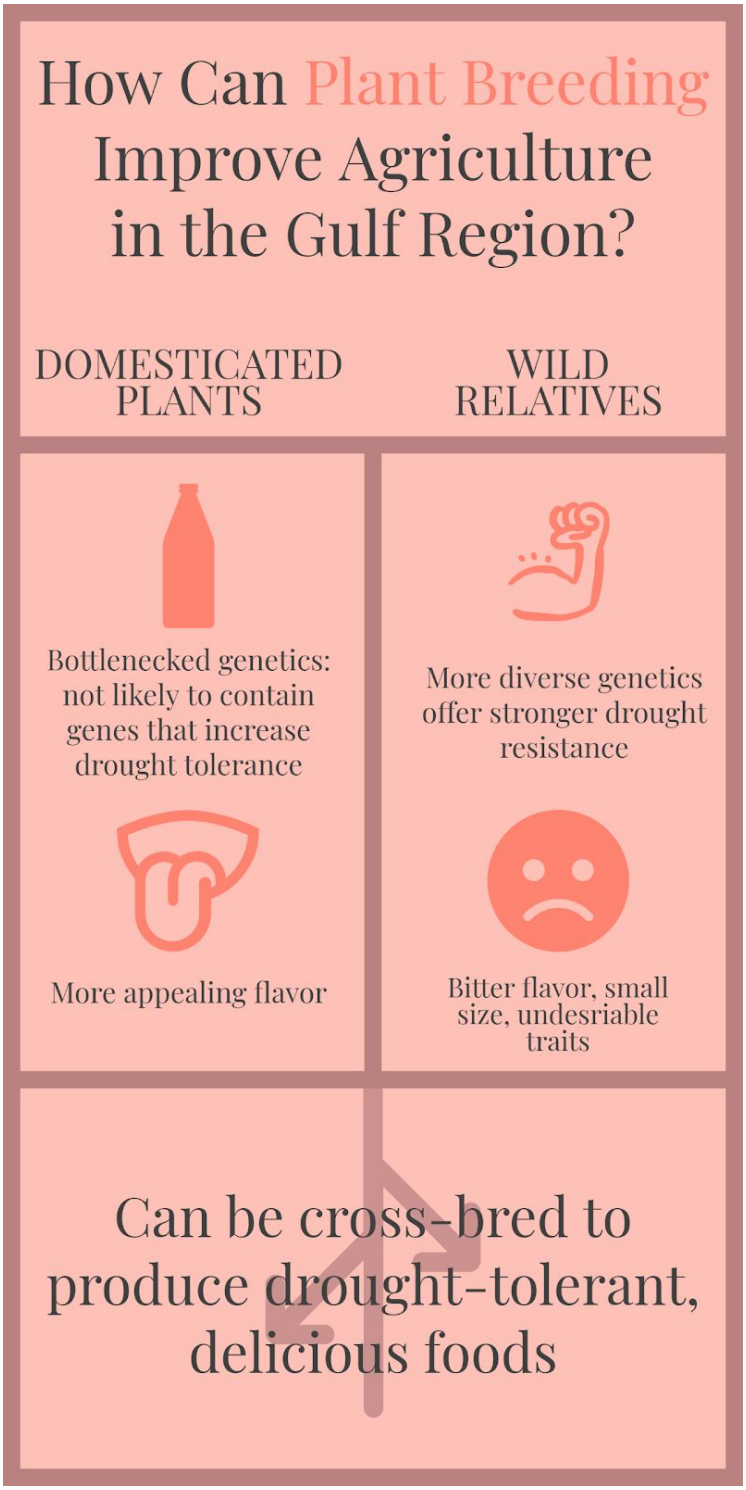

Figure E: Informational graphic about plant breeding and crossing domesticated varieties and wild relatives

Risks in Food Trade and Food Security

In 2017, the World Economic Forum (WEF) released a document, "Shaping the Future of Global Food Systems,"[21] where four scenarios were presented that represent possible futures for the global food system depending on future developments. As a variable of trade connectivity (i.e.the degree to which international food trade takes place) and resource efficiency (i.e. whether recycling, utilization of environmentally-friendly energy and transport practices become more prominent) , these four scenarios for the global food market would each have different impacts on the food market of Bahrain.

For Bahrain, two scenarios that would produce the most favorable results for the population would be the top-left, in a world of high connectivity and resource-intensive consumption, or the bottom-right, in a world of low connectivity and resource-efficient consumption.

In a high-connectivity and resource-intensive environment, Bahrain would effectively be able to continue developing industry without any changes. It would remain dependent on oil and aluminum production, and could continue to primarily source its food supply through imports, as funded by the lucrative oil industry. However, as climate change necessitates a move towards a less resource-intensive world economy, and the oil reserves of Bahrain are limited to a very small region, this may not be effective in the long-term.

In a low-connectivity and resource-efficient environment, Bahrain could substantially fortify domestic food production while simultaneously diversifying the economy. Although described as low-connectivity, connection with other countries could continue through information and finance-based industries, while the food supply would need to be developed using innovative methods such as hydroponics, greenhouses, and through cultivation of crops that grow favorably without climate control.

The ideas behind this graphic, however, do simplify the reality of food production for Bahrain to an extent. Food production and trade is impacted by many factors such as the threat of food import disruption, international supply-chain disruptions, and changing food preferences that favor crops that grow more successfully abroad than in Bahrain.

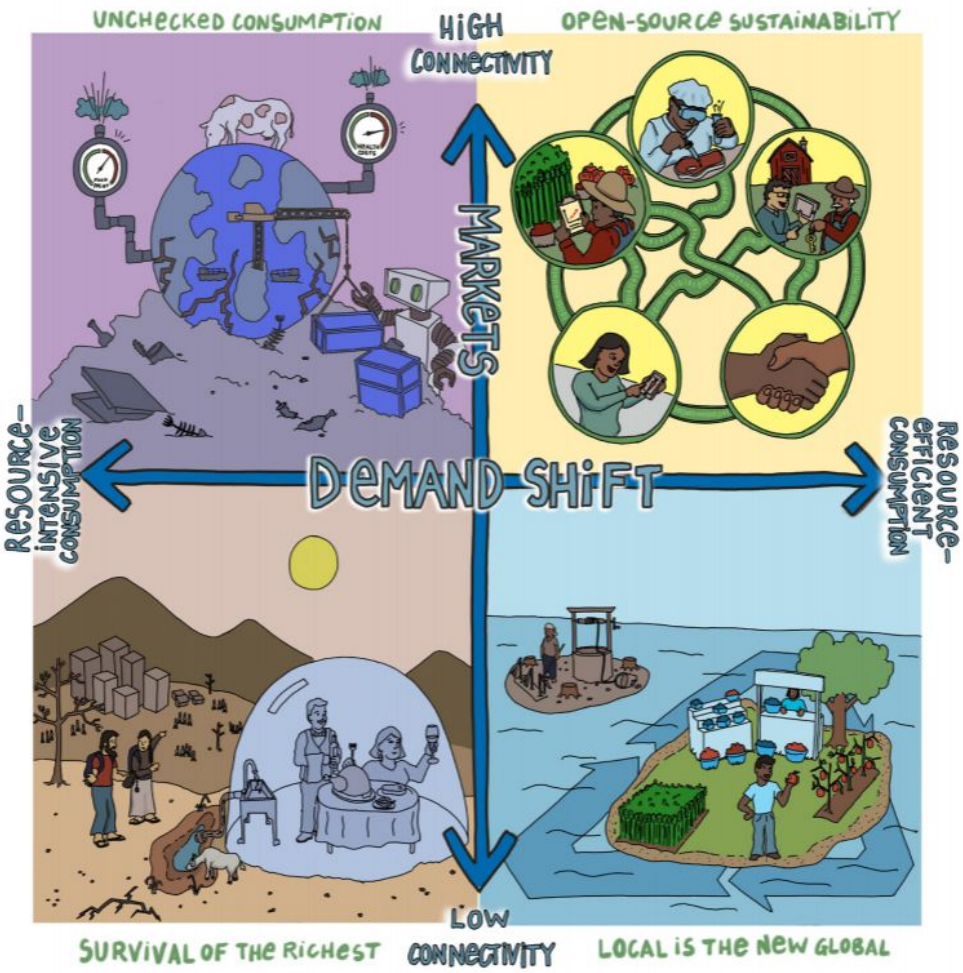

Figure F: Predictions from the World Economic Forum 2017 Report for the future of global food trade

Risk Assessments

Bahrain's dependence on a primarily imported food supply presents some risks to food security for the country. Risks include sudden deterioration of diplomatic relations with trade partners and unreliability of food quality. A major example of how food trade can be unexpectedly jeopardized is the 2017 Qatar diplomatic crisis, where Qatar's food trade partners enacted an embargo that threatened the country's access to items such as vegetables and dairy products. In response to the still ongoing embargo, Qatar has adapted its domestic food production to largely meet demands in some key ways. In other ways, though, Qatar is still dependent on imports as new trade partners were sought out for foods such as grains, and Qatar

continues to depend on imports for livestock feed to support its newly expanded domestic dairy production.[22]

A directly similar situation is unlikely for Bahrain, due to its very close relationship with Saudi Arabic and other Gulf nations, but worldwide supply-chain disruptions like what was experienced in the beginning of 2020 in response to the pandemic caused by the SARS-CoV-2 virus can disproportionately impact Bahrain. Based on the limited impact that the pandemic had on supply-chains as of May 2020, Bahrain's focus was on distributing food to the country's own lease food-secure group: the over 50% of the population who are migrant workers. Food distribution programs came under scrutiny, though, as they were seen to potentially increase spread of the SARS-CoV-2 virus.[23] Notably, though, during the pandemic Bahrain has announced implementation of blockchain technology, or an open-access record of food import transactions, to log import data and food sourcing information in a way that encourages maintaining a stable food supply and may help prevent food shortages.[24] Results of implementing that technology in supply chains, though, may need more time to present benefits.

Changing food consumption patterns present both risks and barriers to developing sufficient domestic food production. If societal tastes change to depend solely on varieties of food that are exclusively successful growing abroad, Bahrain will need to invest more heavily in agricultural technologies that strictly control the growing conditions of food to produce the same varieties locally, such as climate-controlled greenhouses. To mitigate those problems, national programming to improve public perception of traditional crops and drought-resistant varieties could be implemented.

Conclusion

The list of circumstances that work against Bahrain are many: a large population living on a desert island where the economy largely depends on a handful of resources. But despite these challenges, Bahrain has managed to evolve into a well-developed country with incredible access to resources within a single generation. There are risks associated with the current food supply chain in Bahrain, but through identifying risks, solutions can be uncovered that will allow

Bahrain to source food through more stable processes, especially in domestic production. Food systems, though, are complicated. The ideas presented in this paper should be considered recommendations for future developments and not exact answers to the concerns that Bahrain faces. Although these recommendations may not prove successful, it is important to find practical solutions to the question of food import reliance in Bahrain, especially as the worldwide food trade and the world economy continue to change in the next decades.

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